

DOCUMENT 1 — SmartCare v0.1 Initial Engineering Brief and AI Activity Card

SmartCare v0.1 — Initial Engineering Brief

1. Problem Summary

Smart Care Community Clinic are currently using spreadsheets and paper records for patients and appointments. This is very inefficient which are very likely to cause problems for incorrect information or conflicts of appointments.

The software needs to help manage patients, practitioners, and appointments. The system should make everything easier for the staff as then records and access of appointment details are easier to obtain and read.

2. Initial Stakeholders

Stakeholder	Possible Need
Receptionists	Create, view, change, cancel appointments
Practitioners	Appointments and availability
Patients	Information recorded correctly
Clinic Management	View and manage information

3. Initial Features

Feature	Confirmed or Provisional?	Why?
Manage patients	Confirmed	System needs to manage patients
Manage practitioners	Confirmed	System needs to manage practitioners
Manage appointments	Confirmed	System needs to manage appointments
Cancel appointments	Provisional	Not clearly confirmed how cancellations should work
Check practitioner availability	Provisional	Useful but more information is needed

Prevent double bookings	Provisional	Help avoid appointment conflicts but needs to be confirmed
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4. Questions for the Client

1. What patient information needs to be stored?
2. What practitioner information needs to be stored?
3. How long should each appointment be?
4. Should the system prevent practitioners from being booked twice at the same time?
5. Who should be allowed to create, change or cancel appointments?

5. What We Do Not Yet Know

1. What information needs to be stored for each patient and practitioner.
2. What rules should apply when creating, changing or cancelling appointments.
3. What access and permissions different users should have.

The brief specifically says the client's initial statement is **not yet a complete specification**, which is why the provisional features and client questions are important.

AI Activity Card — Ask, Check, Explain

Before AI

What do I think the code does?

The code should create and store appointment info like the patient name, the practitioner and time of appointment.

What problems can I already identify?

The code may accept incorrect or blank information. It may also allow two patients to be booked with the same practitioner at the same time.

AI Request

I asked AI to act as a tutor and explain the code, identify potential problems and help me understand another way of creating the appointment system.

Evaluate

AI Suggestion	Useful	Unclear	Incorrect	Out of Scope
Check for blank patient names	Yes			

Validate appointment information	Yes			
Prevent double bookings	Yes			
Add a database				Yes

Decide

Suggestion	Decision	Reason
Check for blank patient names	Accept	A patient name should be required
Validate appointment information	Keep unverified	More requirements are needed
Prevent double bookings	Keep unverified	The client needs to confirm the booking rules
Add a database	Reject	It is outside the scope of the current prototype

Verify

I verified the AI suggestions by:

- Running the code in PyCharm.
- Testing a normal appointment.
- Testing blank information.
- Testing unusual information.
- Comparing the results with the requirements.

Explain

I can explain how the program asks for appointment information and stores the patient name, practitioner name and appointment time.

I still need to understand more about input validation and how the system could prevent double bookings. I would also need more information from the client before adding features that have not yet been confirmed.